

Ing. Michal Štěpán

Senior GenAI Engineer · LLM applications, evaluation, MCP · Azure & GCP

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Portfolio mxstex.github.io/MyWebPage

PROFILE

GenAI engineer who builds LLM applications end to end. At ABB I work on an AI-assisted validation app, where an LLM works alongside deterministic rules and every change is measured against golden datasets, and I deliver it to Azure Container Apps. I also build MCP servers that give AI coding assistants access to team tools. In my own time I build the same way on GCP as Tojin Games: LLM agents on local models, physics simulation games, a shared login service and an automated video pipeline, developed with Claude Code. Before ABB I spent three years as a freelance data and backend engineer (DTSE, Landis+Gyr, Vendavo) and built energy analytics at Siemens and ABB. Before that, eight years at a district-heating utility, the last ones as technical deputy leading more than fifteen people. I studied nuclear engineering at ČVUT FJFI and did tokamak research there. I want to keep doing this work (designing and shipping LLM applications and agents), with a larger share of the design and technical lead.

CORE SKILLS

GenAI & LLM applications

Azure OpenAI / Azure AI Foundry · structured outputs · LLM + deterministic-rule validation · golden datasets & evaluation · Model Context Protocol (MCP) servers · LLM agents on local models (Ollama) · pgvector memory · AI coding agents (Claude Code, GitHub Copilot)

Programming & testing

Python (FastAPI, Pandas, NumPy/SciPy, NiceGUI) · Pytest · Playwright · load and stress tests · SQL / PostgreSQL · JavaScript (Canvas, WebGL2) · Kotlin + Jetpack Compose

Cloud & delivery

Azure Container Apps · Container Registry · Key Vault with managed identity · Log Analytics · Blob & Table Storage · Azure Pipelines (dev → prod) · GCP Cloud Run · Secret Manager · Terraform (GCP) · Docker · OAuth 2.0 / JWT

Worked with earlier

Dagster · Airflow · DuckDB · BigQuery · RabbitMQ · Kubernetes · GitLab CI · GitHub Actions · Flask · Streamlit · AWS (S3, Lambda) · Azure Functions · Azure AI Search · Document Intelligence · LangChain · LangGraph · ChromaDB · Grafana · Power BI · ARIMA / SARIMA

PROFESSIONAL EXPERIENCE

Senior GenAI Engineer · ABB s.r.o.

Sep 2025 – present

- AI-assisted validation of engineering spreadsheets: built the evaluation (golden datasets, acceptance evaluation, performance and stress tests) that every change is measured against.
- Backend in FastAPI: a YAML rule engine that runs deterministic checks next to the LLM validators, switchable model profiles (Azure OpenAI or a local model through Ollama) with structured outputs, quotas and session cleanup; benchmarked local against cloud models.
- Delivery: Azure Pipelines for dev, test, stage and prod; Docker images in Container Registry, Azure Container Apps, Key Vault secrets through managed identity, Log Analytics.
- Built a proof-of-concept MCP server (Python, FastMCP) that gives AI coding assistants (Claude Code, GitHub Copilot) access to Azure DevOps work items, with a mock API and tests.
- Wrote the architecture documentation for a RAG assistant (Azure AI Search hybrid retrieval, LangGraph, Azure OpenAI).

Stack: Python · FastAPI · Azure OpenAI / AI Foundry · Pandas · Pytest · Docker · Azure Container Apps · Key Vault · Azure Pipelines · MCP · Ollama

Freelance Software / Data Engineer

Sep 2022 – Sep 2025

DTSE, Landis+Gyr, Vendavo and other clients in energy, telecom and manufacturing · remote, international teams

- Designed and built end-to-end data workflows in Python, orchestrated with Dagster and Airflow, stored in PostgreSQL, DuckDB and BigQuery.
- Built backend components for high-volume ingestion and message-based processing with RabbitMQ, and REST services.
- Migrated services from Kubernetes to cloud-native GCP (Cloud Run, GCS, BigQuery).
- Maintained CI/CD in GitLab CI and GitHub Actions with unit, integration and end-to-end tests in Pytest.
- Delivered high-performance processing pipelines for recommender systems (multiprocessing, optimisation); L2 support for production systems.

Stack: Python · Dagster · Airflow · PostgreSQL · DuckDB · BigQuery · RabbitMQ · GCP · Kubernetes · GitLab CI · GitHub Actions

Senior Software Developer · Siemens s.r.o.

Jan 2021 – Aug 2022

- Developed an Energy Management System consolidating and analysing data from many sources.
- Implemented ARIMA and SARIMA models for time-series forecasting.
- Mentored junior developers.

Stack: Python · Flask · FastAPI · Docker · PostgreSQL · Grafana · Power BI · RabbitMQ · AWS S3 · AWS Lambda

Software Project Engineer · ABB s.r.o.

Oct 2018 – Jan 2021

- Developed a water and energy consumption evaluation system; designed and implemented predictive-analytics features.

Stack: Python · Docker · PostgreSQL · Azure Functions · Azure Blob Storage · Grafana · Power BI

IT Specialist · Power Engineer · Technical Deputy · DISTEP a.s.

May 2010 – Oct 2018

- As technical deputy, led an operations team of more than fifteen people in water and heat distribution.
- Improved resource efficiency through optimised calculations and monitoring; streamlined data collection with PDA barcode readers (MS SQL, VBA).

TOJIN GAMES · OWN STUDIO, IN MY OWN TIME

Independent science, space and simulation games · YouTube youtube.com/@tojin111 · live builds on the portfolio site

- Built eight science, space and strategy games solo, plus an early online prototype of Jednota; five are live in the browser on Google Cloud Run. Hand-written simulation engines for Newtonian gravity, Kepler orbital mechanics with WebGL2 rendering, plasma confinement (Biot–Savart) and the Schrödinger equation (split-step Fourier). In Gravity, tests keep the browser JavaScript engine bit-identical to the Python reference and the Kotlin Android engine within 1e-9.
- **Jednota**, a 4X strategy game run by LLM agents (~73,000 lines of Python, about 1,300 tests): strategist, envoy and fleet-commander agents on local LLMs through Ollama, with JSON-schema output and no agent framework; deterministic rules validate every model decision (accepted 93/93 strategic, 80/88 diplomatic, 79/86 tactical in a measured five-empire campaign); memory across games through pgvector RAG.
- **Platform**: an OAuth 2.0 to RS256 JWT entitlement service (FastAPI, Cloud Run, Secret Manager, Terraform, 54 tests) gives five games one shared login and access tiers; every service scales to zero under a one-instance cost cap.
- **Video pipeline**: code-driven 4K capture of real game sessions (Playwright, Chrome DevTools Protocol, ffmpeg/NVENC), captions from on-screen events, local TTS, loudness-checked masters and automated validation, plus local ComfyUI models (FLUX, Wan 2.2): more than forty videos published in September 2026.
- Developed with Claude Code: about 70% of the commits are co-written with the agent. The results are checked by test suites, level verifiers and bots that play the games through.

EDUCATION, CERTIFICATES, LANGUAGES

Ing. (MSc), Electrical Engineering and Computer Science · 2012

VŠB – Technical University of Ostrava. Thesis: VBA Utilization for Power Engineering Data Processing and Presentation

Bc., Nuclear Engineering · 2009

Czech Technical University in Prague, Faculty of Nuclear Sciences and Physical Engineering. Thesis: Physical Analysis of Fast Reactor Cores in Up-to-Date Tendencies

Research project · 2006

Institute of Plasma Physics, Czech Academy of Sciences: electrostatic turbulence in the edge plasma of the CASTOR tokamak

Microsoft AZ-305

Designing Microsoft Azure Infrastructure Solutions · training

Microsoft AZ-400

Designing and Implementing Microsoft DevOps Solutions · training

Cambridge English: First (FCE)

Grade B · CEFR B2 · 2019

Czech

native

English

B2–C1 (Cambridge FCE)